

Handling Mixed Variables

Module 33 Study Notes • Feature Engineering

1. What is Mixed Data?

In Machine Learning, real-world data is rarely pristine. Sometimes you will encounter a single column in your dataset that contains a mixture of both **numerical data (numbers)** and **categorical data (strings/text)**. This is called a Mixed Variable, and it will instantly break any standard Scikit-Learn transformer (like `SimpleImputer` or `OneHotEncoder`) if you try to process it as-is.

Mixed data typically manifests in one of two distinct patterns:

2. Type A: Sequential Mixed Data

In this pattern, a single column contains discrete numbers in some rows, and discrete text strings in other rows.

- Row 1: **5**
- Row 2: **3**
- Row 3: **'Alone'**
- Row 4: **'Group'**

The Strategy:

You cannot pass this to a model. You must split this single column into **two entirely new columns**: one exclusively for the numerical values, and one exclusively for the categorical values. Where a number exists, the categorical column receives a `NaN`. Where a category exists, the numerical column receives a `NaN`.

```
import pandas as pd
import numpy as np

# 1. Create the Numerical Column
# pd.to_numeric forces strings to NaN (errors='coerce')
df['number_col'] = pd.to_numeric(df['mixed_col'], errors='coerce')

# 2. Create the Categorical Column
# Keep the original value if it is NOT a number, otherwise set to NaN
df['category_col'] = np.where(df['number_col'].isnull(), df['mixed_col'],
```

`np.nan)`

3. Type B: Fused Mixed Data (Alphanumeric)

In this pattern, the numerical and categorical information are fused together inside a single string within the same cell.

Example (Titanic Cabin Column):

- `'C85'` (Deck C, Room 85)
- `'B22'` (Deck B, Room 22)

If you treat 'C85' and 'C86' as unique categories, you will have hundreds of unique categories (High Cardinality) and lose the broader pattern (they are both on Deck C). The information must be decoupled.

The Strategy:

You must use **Regular Expressions (Regex)** or Python string slicing to extract the text part into one column and the numeric part into another column.

```
# 1. Extract the Numeric Part
# The regex (\d+) looks for one or more digits and extracts them
df['cabin_num'] = df['Cabin'].str.extract('(\d+)')

# 2. Extract the Categorical Part
# The regex ([a-zA-Z]+) looks for one or more letters and extracts them
df['cabin_cat'] = df['Cabin'].str.extract('([a-zA-Z]+)')

# Now you can One-Hot Encode 'cabin_cat' and Impute/Scale 'cabin_num'!
```

4. Summary

Whenever you encounter a mixed column, your primary objective is decoupling. The machine learning pipeline requires homogeneity. Once you split the mixed column into a pure numerical column and a pure categorical column, you can pass them into your standard `ColumnTransformer` pipeline for imputation and encoding just like any normal dataset.